

Quotient Laws of Exponents

Lesson Plan

1. Lesson Information

Course	Grade 7 Accelerated (1205050)	Unit	3
Lesson	Lesson 2 — Quotient Laws of Exponents	Date	_____
Length	53 minutes (one period)	Benchmark(s)	MA.7.NSO.1.1

2. Standards

MA.7.NSO.1.1 Know and apply the laws of exponents to evaluate numerical expressions and generate equivalent numerical expressions, limited to whole-number exponents and rational-number bases.

Must. Whole-number exponents with rational bases; the zero exponent is derived from the quotient law, not announced.

Must not. Bases are nonzero wherever a quotient or a zero exponent appears, and 0^0 is never asked. Negative exponents wait for 3.04.

3. Mathematical Thinking and Reasoning Standards

the evidence for each, from this period

MTR	What it asks	The evidence in this lesson
MA.K12.MTR. 5.1	Use patterns and structure to help understand and connect mathematical concepts.	Notes III — the class reaches 2^0 two ways, by cancelling matching factors and by subtracting the exponents, and sees that the value 1 is forced by the law rather than declared.
MA.K12.MTR. 6.1	Assess the reasonableness of solutions.	Board 3 — $(-6)^6$ over $(-6)^5$ leaves one factor of -6 ; students say whether the answer is positive or negative before they write anything.
MA.K12.MTR. 4.1	Engage in discussions that reflect on the mathematical thinking of self and others.	Board 5 — when the room splits between 1 and 0 on 9^0 , the sixty-second re-vote before the reveal.

4. Learning Target and Essential Question

Learning target. I can use the quotient of powers, power of a quotient and zero exponent laws to rewrite and evaluate a numerical expression.

Essential question. Why must the bases be the same to apply the quotient of powers law?

Building on. Lesson 3.01 — the product laws; MA.6.NSO.3.3 — evaluating powers.

Working toward. MA.8.NSO.1.3 — the same laws with negative exponents (3.04–3.07), where $2^3/2^5 = 2^{-2}$ is the step after today's $2^5/2^3 = 2^2$.

5. Sequence

read from the deck; never typed

Segment	Min	What the teacher does	What students do
Settle in; post the learning target	1	Post the target; say the 'today' line and nothing else yet.	Copy the target.
Warm-Up	5	Two minutes silent, then reveal. One sentence of reteach per question at most.	Four retrieval questions, alone, no notes.
Notes	11	Teach from the front. Expand once per law; say the OFF line out loud.	Copy the notes; answer the checks aloud.
Example 1	3	Work the example; name the wrong boards before they appear.	Watch, then work the Your Turn alone.
Your Turn	3	Circulate; reveal and name what a wrong board most likely did.	Work it alone; boards up on the cue.
Example 2	3	Work the example; name the wrong boards before they appear.	Watch, then work the Your Turn alone.
Your Turn	3	Circulate; reveal and name what a wrong board most likely did.	Work it alone; boards up on the cue.
Whiteboards	13	One question at a time. Boards down until the cue, all up together; scan the back row first.	Answer on a board; hold it up on three.
Independent Set	6	Silent. Circulate and mark what you see; do not teach.	Six questions, written, alone.
IXL	5	Assign the day's IXL skills.	Start the skills; finish tonight.
Total	53		

6. Gradual Release

I do. The Notes slides and the worked half of each Example: the teacher writes, the class copies, and the OFF line is said out loud rather than printed.

We do. Each Example's Your Turn: the same steps on the student's own numbers, revealed and named a minute later.

You do. The nine-question whiteboard round (13 minutes) and then the six-question independent set, silent and written.

7. Higher-Order Questions

Question	DOK
Cancelling $2^3/2^3$ gives 1 and subtracting the exponents gives 2^0 . Why must those be the same number?	DOK 3
The quotient law says the base has to be nonzero. Say what goes wrong if it is 0.	DOK 2
$0.5^8 \div 0.5^5$ is 0.125, which is smaller than 1. Explain how dividing one power by another can leave a number under 1.	DOK 3

8. Checks for Understanding

each board, and what to do about it

#	What the board asks	Answer	If the board is wrong
1	$\frac{10^5}{10^3}$	10^2	10^8 — added the exponents [confused with product of powers]; 1^2 — divided the bases [7.NSO.1.1 misconception: operating on the bases]
2	A server stores 8^4 files in 8^1 folders, evenly. It has been running for 30 days. How many files are in each folder?	512	4096 — kept 8^4 and treated 8^1 as 1 [identity law misread as zero exponent]; 64 — subtracted $4 - 1$ wrong or used 8^2 ; anything built from 30 — the unneeded figure was used
3	$\frac{(-6)^6}{(-6)^5}$	-6	6 — dropped the sign of the base; 1 — divided the bases [7.NSO.1.1 misconception]
4	$\left(\frac{-5}{10}\right)^7$	$\frac{(-5)^7}{10^7}$	$(-5)^7/10$ — exponent applied to the numerator only [7.NSO.1.1 misconception: exponent applied to part of a base]; $(-0.5)^7$ — simplified first (correct value, but not what was asked)
5	9^0 Which is the value?	A — 1	(B) confused the zero exponent with the identity exponent, 9^1 [7.NSO.1.1 law names]; (C) treated the zero exponent as a reciprocal, which is what a NEGATIVE exponent does [misconception carried into 8.NSO.1.3]; (D) zero exponent read as 'nothing', so 0 [8.NSO.1.3: zero exponent law, $a^0 = 1$]
6	$\frac{4^8}{4^2}$ Which expression is equivalent?	C — 4^6	(A) divided the bases as well as subtracting the exponents [7.NSO.1.1 misconception: operating on the bases]; (B) added the exponents — the product law applied to a quotient [confused with product of powers]; (D) divided the exponents, $8 \div 2$ [7.NSO.1.1 misconception: 'divide' applied to the exponents]
7	$\frac{0.5^8}{0.5^5}$	0.125	0.5^{13} — added the exponents instead of subtracting [quotient of powers confused with product of powers]; 0.5^{40} — multiplied the exponents [confused with power of a power]
8	$\left(\frac{10}{11}\right)^0$	1	$10/11$ — identity exponent instead of zero exponent; 0 — zero exponent read as 0 [8.NSO.1.3: zero exponent law, $a^0 = 1$]
9	Hannah cancels $2^3/2^3$ to 1. Sebastian subtracts to 2^0 . Who is right?	Both — $2^0 = 1$	'Hannah, because $2^0 = 0$ ' — zero exponent read as 0 [8.NSO.1.3: zero exponent law, $a^0 = 1$]; 'Sebastian only' — did not accept that a nonzero number over itself is 1

MTR.4.1 — the one peer move. When a board splits — no option holding about two-thirds of the room — sixty seconds of “convince the person next to you”, then re-vote, then reveal. The minute comes out of the whiteboard remainder.

9. Differentiation

ESE / IEP. Write the quotient expanded once — $8 \cdot 8 \cdot 8 \cdot 8$ over 8 — and leave that line on the desk. On the first two boards the student crosses out matching factors and counts what is left; subtraction of exponents takes over from board 3.

ELL. Quotient and product are the two words in play. Put QUOTIENT = divide and PRODUCT = multiply on the board with $8^4 \div 8^1$ under one and $8^4 \cdot 8^1$ under the other, and point at them as each board goes up.

Enrichment. Ask why 0^0 is left undefined while 9^0 is 1, then for the value of $(2/3)^5 \div (2/3)^5$ without computing either power.

10. Closure, Vocabulary, Materials, Practice

Closure. Board 9 is written work and it is the exit evidence: the student has to say that both students are right and that 2^0 is 1. Read the boards, not the papers.

Vocabulary. **quotient of powers law; power of a quotient law; zero exponent law**

Materials. Whiteboards and markers. Calculators are allowed on everything; $5^6 = 15,625$ is on the notes for the value check.

Practice. The six-question independent set is worked silently in class; whatever is not finished goes home. IXL, all skills required to a SmartScore of 67, due at the start of the next class: Powers with negative bases — WFW; Division with exponents — XNC; Identify equivalent expressions involving whole-number exponents — KBE.